

5.4.4

EE25BTECH11009 - Anshu Kumar Ram

Question:

Using elementary transformations, find the inverse of the following matrix

$$\begin{pmatrix} 2 & 4 \\ -1 & 2 \end{pmatrix}$$

Solution:

We know that

$$\mathbf{A}^{-1}\mathbf{A} = \mathbf{I} \quad (0.1)$$

where $\mathbf{I}$ is the 2×2 identity matrix.

Now we form the augmented matrix:

$$\left(\begin{array}{cc|cc} 2 & 4 & 1 & 0 \\ -1 & 2 & 0 & 1 \end{array} \right) \xrightarrow[R_2 \leftarrow R_2 + \frac{1}{2}R_1]{R_1 \leftarrow \frac{1}{2}R_1} \left(\begin{array}{cc|cc} 1 & 2 & 0.5 & 0 \\ 0 & 3 & 0.5 & 1 \end{array} \right) \quad (0.2)$$

$$\xrightarrow[R_1 \leftarrow R_1 - \frac{2}{3}R_2]{R_2 \leftarrow \frac{1}{3}R_2} \left(\begin{array}{cc|cc} 1 & 0 & \frac{1}{3} & -\frac{2}{3} \\ 0 & 1 & \frac{1}{6} & \frac{1}{3} \end{array} \right) \quad (0.3)$$

Therefore,

$$\mathbf{A}^{-1} = \begin{pmatrix} \frac{1}{3} & -\frac{2}{3} \\ \frac{1}{6} & \frac{1}{3} \end{pmatrix} \quad (0.4)$$

This can be verified from the code by showing that

$$\mathbf{A}^{-1}\mathbf{A} = \mathbf{I} \quad (0.5)$$